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Development and clinical application of a probabilistic robustness evaluation tool for pencil beam scanning proton therapy treatments

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A B S T R A C T

Purpose: to implement a probabilistic-Robustness-Evaluation (pRE) tool for proton therapy treatments and to correlate these results with the worst-case approach (wRE) implemented in commercial TPS for clinical applications.
Materials and Methods: 12 skull base patients were planned with a robust multiple field optimization (MFO) approach. 10 years of machine QA were analysed to derive the uncertainties of our treatment system (beam delivery and patient positioning system). For a large cohort of patients, post-treatment imaging was acquired to determine the intra-fraction uncertainty. The pRE, considered explicitly all these uncertainties, the fractionation and range uncertainty. For each plan a wRE with different combinations of range and setup uncertainties was simulated. wRE results were then compared, in terms of target coverage and OAR dose limits, with pRE results.
Results: 43,400 dose distributions were analysed. pRE simulations lasted 18.6 h (±11.5 h). The results showed that the combination of wRE uncertainty parameters that surrogated the best pRE results with a confidence level of 95 % were (1.0 mm/3.5 %). The median OAR's dose indexes difference (D_1/D_{1cc}) between pRE and wRE was 1.90 (±1.49) GyRBE, while for target D_{98} and D_{95} it was −0.66(±0.95) and −0.67 (±0.52) GyRBE, respectively.
Conclusion: A tool able to explicitly simulate the source of treatment uncertainties and the effect of the fractionation was implemented to have a more realistic evaluation of plan robustness. This tool was used to find the best wRE parameters that surrogate the pRE results while maintaining clinically acceptable timing. These results are now used in our clinical workflow.

1. Introduction

Every high conformity radiation therapy technique (either for photons and protons) is prone to dose distribution variations due to uncertainties (e.g. setup, mechanic, dosimetric etc.) that can have a significant impact on clinical dose distributions. Proton therapy (PT) has the additional source of uncertainty coming from the proton's unique physical characteristic of having a finite range in matter. The determination of Bragg peak's range is related to the accurate and consistent determination of tissue parameters for range prediction from computed tomography scans, namely the stopping-power ratio, the dose calculation algorithm, the planning-CT grid size etc [1]. Another source of uncertainties directly correlated to particle therapy is the radiobiological effectiveness (RBE) of such radiation and its dependence on dose, LET, tissue and patient factors [2]. The high sensitivity of plans to composite uncertainties increases the risks of underdosing the target or overdosing organs at risk (OARs) [3]. The availability of a reliable tool to verify the robustness of plans with respect to different sources of uncertainties is crucial to guarantee the quality of the treatments.

A recent survey from Kaplan et al. [4], shows the variety of methodologies and metrics for robustness evaluations used in clinical workflows. In general, two types of approaches are used to evaluate plan robustness in proton therapy [5]. The first, from here on named *worst-case scenario* approach, considers all the errors as systematic and does not consider joint probabilities of uncertainties occurrences. Moreover, the mitigation effect of the fractionation and of random errors is taken into account only if the setup uncertainties are simulated by using the van Herk's formula in which the random errors are simulated by scaling the systematic errors by a factor of 0.7. Many papers proposed different solutions with this approach but all of them considered the setup uncertainty as the unique surrogate of all uncertainties except for the range uncertainty that was explicitly simulated [6–12]. This approach has the

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advantage of low computational costs which makes such methods usable in clinical workflows. The con is related to the fact that an improper choice of uncertainty parameters can lead to a wrong estimation of the effect of such errors on clinical dose distributions. So far, this method is the only one implemented in commercial solutions [7,13].

The second approach, from here on named *probabilistic* approach, is based on sampling errors from their distribution with a large number of dose scenarios simulated [14–19]. The proper number of scenarios can be estimated by analysing the statistical significance of the results in the dosimetric space [16,20]. The impact on calculation time, which is the most limiting factor of these methods, can be reduced by using analytical dose engines [16,18] or by using variance reduction techniques [21]. Lately deep-learning based approaches have been developed to predict plan robustness by using probabilistic methods at very low computation times [22].

In this study we present the implementation of a probabilistic robustness evaluation (pRE) tool based on ten years of machine QA and patient treatment data collected at our proton therapy centre to simulate the effect of uncertainties on treatment plans; fractionation is also taken into account. To the best of our knowledge this is the first study wherein a probabilistic robustness evaluation approach uses real data derived from clinical activity and a Monte Carlo dose calculation algorithm with no analytical approximations. The results of the analysis performed with the probabilistic approach were compared with the results of robustness analysis obtained with the worst-case scenario approach (wRE) implemented in our commercial TPS considering with different systematic error values to find the best parameters' combination able to surrogate the pRE results.

2. Materials and methods

2.1. Clinical cases

12 patients with a base of skull chordoma or chondrosarcoma were selected for this study. In Table 1 dose prescription of the high-risk volume and the number of fractions is reported. The clinical request for nominal target coverage was $D_{95} = 95$ % of dose prescription. In Table 1, constraints to primary OARs are reported. These are defined as primaries because they are the only limit for target coverage. In some cases, these dose values have been violated or lowered due to clinical consideration of patient comorbidity or high risk of acute toxicity. In particular, this study is focused on target coverage (in terms of $D_{98\%}$ and $D_{95\%}$) and dose to only primary OARs such as brainstem, chiasm, optic nerves and posterior optical pathways (in terms of D_{1cc} for the brainstem and $D_{1\%}$ for the others). Each patient was positioned on the treatment couch by using a three fixation points thermoplastic mask.

Table 1
Clinical cases. Dose to high-risk volume and number of fractions (above) and dose constraints to primary OAR (below) are reported.

District	**Number of patients**	**Dose prescription** Median (min/max) [GyRBE]	Number of fractions Median (min/max)
Base of skull	12	72 (72/74)	36 (35/40)
		OAR constraints	
Brainstem		$D_{1cc} < 60$ GyRBE	
Chiasm		$D_1 < 54$ GyRBE	
Optic Nerves		$D_1 < 54$ GyRBE	
Posterior optical pathways		$D_1 < 54$ GyRBE	
Spinal cord		$D_1 < 50$ GyRBE	
Eyes		$D_1 < 50$ GyRBE	
Temporal Lobes		$V_{60\ GyRBE} < 5.5$ cc $V_{70\ GyRBE} < 1.7$ cc	
Lacrimal Glands		$D_{mean} < 24$ GyRBE	
Cochlea		$D_{mean} < 35$ GyRBE	
Oesophagus		$D_{mean} < 34$ GyRBE	

2.2. Planning strategy

The TPS used for creating the plan was RayStation (v12A). Every plan was robustly optimised, with a minimax approach, on the CTV considering a setup error of 2.0 mm in every cardinal direction and a range uncertainty, modelled as an HU to stopping power curve miscalibration, of ± 3.5 %: 21 scenarios were considered during the optimization. Target coverage and maximum objective functions to primary OARs (such as brainstem, optic chiasm, optic nerves, posterior optical pathways and temporal lobes) were defined to be optimized in the nominal and in the error dose scenarios. Secondary OARs (such as pituitary gland, hippocampus, oral cavity etc.) objective functions were defined in order to be considered only in the nominal dose scenario.

Four non-coplanar beam directions were used. For every beam entrance direction, when needed, the range shifter optimization technique was adopted to limit its use in the shallow region of the target and to reduce the lateral penumbra in depth [23]. For every patient a dose grid resolution of 2.0 mm and a statistical uncertainty of 1.0 % was set. The dose calculation algorithm used was the Monte Carlo (v5.4) running on GPU [24].

2.3. Uncertainties estimation

To consider all the machine related sources of uncertainties that can occur during the treatment, yearly geometric and dosimetric QA performed in two gantry rooms were reviewed from 2014 until today. Moreover, we evaluated the intrafraction accuracy of the positioning and immobilization workflow employed in 157 patients with brain tumours at our particle therapy centre. This extensive review of ten years of clinical activity was the key to have the best level of confidence in describing the distribution of every uncertainty involved and to have a high level of confidence in the dosimetric space where robustness analysis was performed. Once the distribution of each uncertainty was derived from this analysis, the mean and the variance values were used to simulate a normally distributed uncertainty for each of these errors. Hereafter is the list and the detailed description of each uncertainty simulated.

2.3.1. Gantry uncertainties

The uncertainties related to gantry rotation were of two types. The first one was the "gantry rotational uncertainty": the nominal gantry angle requested could not be the actual beam delivery direction both because of scanning magnets settings and because of mechanical uncertainty due to gantry inertial mass (hundreds of tons). The QA we performed to evaluate this type of uncertainty was the so-called star shot [25]. For 36 gantry angles (at steps of 10°) a central spot was delivered on a radio chromic film previously aligned with the X-ray imaging system to the isocenter. The difference between the nominal gantry angle and the beam angle measured on the radiochromic film was registered.

The second type of gantry related uncertainty is the "gantry isocentricity": the position of the isocenter, as seen from the gantry eye view, changes as a function of the gantry angle. This happens because such a heavy and large machine could not be perfectly isocentric during the rotation. This test was performed by placing a radiopaque marker at the isocenter by using the X-ray imaging system and acquiring two orthogonal X-ray images for 36 gantry angles (10° step). The shift of the marker, as seen from each gantry angle, was registered along the three cardinal directions.

Both of these types of uncertainties were determined and simulated as a function of the gantry angle.

2.3.2. Couch uncertainties

As in the case of the gantry, the treatment couch could introduce two types of uncertainties during patient positioning. The first one was the "couch rotational uncertainty". This QA was performed, as in the star shot [25] test performed for the gantry, by placing a radiochromic film

on the couch and, delivering a single central spot (gantry in horizontal position), by rotating the couch from −90° to + 90° at 10° steps. The difference between the nominal and the measured couch angle was registered.

The second type of couch related uncertainty was the "couch Isocentricity": the position of the isocenter changed as a function of couch angle since the rotations were not perfectly isocentric because of couch inertia, the mechanical backlash effect at the end of the rotation, the weight of the patient on the couch, the position of the isocenter on the couch reference system (further away is the rotating point from the anchoring point of the couch's robotic arm the higher the uncertainty) and the angle selected (the larger the rotation the higher the uncertainty). This test was performed by placing a radiopaque marker at the isocenter position on the couch at an angle 0°. Two orthogonal X-rays were acquired for each angle (from −90° to + 90° with 10° of step angle). The shift of the marker as it was seen from each couch angle was registered along the three cardinal directions. To evaluate whether or not a dependence on the patient's weight is present, the couch isocentricity uncertainty was measured for three different weights: 20 kg, 60 kg and 100 kg. Both of these types of uncertainties were determined and simulated as a function of couch angle and of the weight loaded on the couch.

2.3.3. Spot position uncertainty

The spot positioning accuracy is one of the QA repeated on a daily, monthly and yearly basis with different detectors and action levels [26,27]. For the purpose of this study, yearly QA were analysed. During such checks the spot positioning was verified by using a scintillating screen detector coupled with a CCD camera [28]. A squared spot pattern of 25 spots was delivered for 12 gantry angles (30° angle step) and 9 energies (from 70 to 226 MeV). The spot positions determined with an in-house developed software were compared (both as a function of gantry angle and beam energy) with the nominal expected positions and the differences, along the X and Y axis in the beam's eye view reference system, were registered.

2.3.4. Range uncertainty

The uncertainty of the proton beam range, intended as the position of the Bragg peak in water as a function of energy have a negligible impact on the total range uncertainty (over the last ten years the average range variation during the yearly QA, over the entire energy interval, was below 0.5 mm). The main contribution comes from the HU to Stopping Power Ratio (SPR) conversion curve reliability and dose calculation algorithm [29]. It is hard to define and quantify each of these contributions. An attempt was made by Paganetti [1] in which a mean value of 2.4 % (1.5 standard deviation) was recommended for PBS planning. A recent study, in which our center was involved, assessed the inter-center variability in stopping power and range prediction within European particle therapy centers [30]. The study showed that in the vast majority of proton centers the Paganetti's hypothesis is still valid and, unless methods to reduce the range uncertainty are applied [31–33], it should be simulated with the value suggested.

2.3.5. Patient positioning: Setup and immobilization device accuracy

To evaluate the setup and intrafraction accuracy of the three fixation points thermoplastic mask, during the first treatment week (fraction 2 to 6), and then once a week until the end of treatment, 157 patient positioning data (head-and-neck, brain and skull base cases) treated at our particle therapy center were analysed. Treatment position verification was accomplished with a gantry-mounted system enabling X-ray imaging. The procedures took place with the following workflow:

1. The patient was positioned by using two orthogonal kV images on bone structures. We corrected every residual error (the patient never started the treatment with a residual error higher than 0.2 mm on average corresponding to the sensitivity of the couch movement).
2. Once the treatment was completed, the couch was moved back to the position approved before the patient's treatment delivery start.
3. Another set of orthogonal kV images was acquired and reviewed by the radiation oncologist. Both translational and rotational corrections were assessed.
4. The 6D correction vectors were registered and stored.

2.4. Robustness evaluations

Once all the uncertainties have been derived from machine QA and patient positioning imaging, a probabilistic robustness evaluation (pRE) tool was implemented (CPython 3.8). The tool was able to calculate in the TPS, with the clinical dose engine (Monte Carlo running on GPU [24]) the plan dose for each fraction of the treatment: as shown in Fig. 1, for each fraction, every source of uncertainty was explicitly simulated from the distribution derived from the detailed analysis of QA results. Some uncertainties were simulated as a function of gantry angles (i.e. gantry isocentricity and rotation and spot positioning), spot positioning was simulated also as a function of beam energy and couch isocentricity and couch rotation uncertainty were simulated as a function of weight load on the couch and couch nominal angle; all of them were simulated by using look up table created on the basis of the above-mentioned QA analysis. All of these dependencies were explicitly simulated in order to take into account any possible systematic errors that can occur during the treatment at certain mechanical/dosimetric parameters selected. The dose calculated for each fraction was summed to obtain the *scenario dose* distribution: from here on intended as the dose of the entire treatment that takes into account both uncertainties and fractionation.

This entire workflow was repeated 100 times to increase the statistical significance of the simulations. This factor has been called *multiplicity* factor and, as in every Monte Carlo method, the higher the number of samples, the more accurate the probability estimated in the dosimetric space will be. With this approach, for each robustness analysis, 100 dose scenarios were simulated. Each scenario was the result of N dose calculation, where N is the number of fraction doses accumulated. This means, for example, that for a patient with a treatment of 35 fractions, 3500 dose calculations were performed and cumulated for each treatment course to finalize the robustness evaluation. The calculation time for each patient was also recorded. The statistical uncertainty of the method was estimated by repeating the pRE with different pseudorandom seeds for one patient.

The version of robustness analysis currently implemented in the TPS had a worst-case scenario approach in which systematic range uncertainties and systematic isocenter shift can be simulated [34,35]. To find if a correlation between probabilistic and worst-case robustness evaluation existed, a wRE was run for every plan with all the possible combinations of setup and range errors with the following values: 2.0 mm, 1.0 mm, 0.5 mm and 0.0 mm for setup uncertainties and 3.5 %, 3.0 %, 2.0 % and 1.0 % for range uncertainty. Each of these analyses consisted of 16 scenarios: 8 isocenter shift in all the diagonal directions times 2 values of range uncertainties. As it can be noticed, these shifts are different from the ones used during optimization because, as for our clinical practice, we are used to evaluate plan robustness on scenarios where the magnitude of the isocenter shift is the same as the one used in the optimization but the direction is different.

3. Dose index comparisons

The study compared the dose indexes of the brainstem, optic nerves, optic chiasm and posterior optical pathways as listed in Table 1. The target coverage was considered based on both D_{95} and D_{98}. From pRE the dose index value equivalent to the 95th percentile, for OARs, and the 5th percentile, for target coverage, of all the scenarios simulated were extracted and compared to the worst-case result of the same dose index in wRE. In Fig. 2 an example of chiasm's D_1 (Fig. 2a) and CTV's D_{98} (Fig. 2b) distribution is shown. In orange the 95th and 5th percentile are

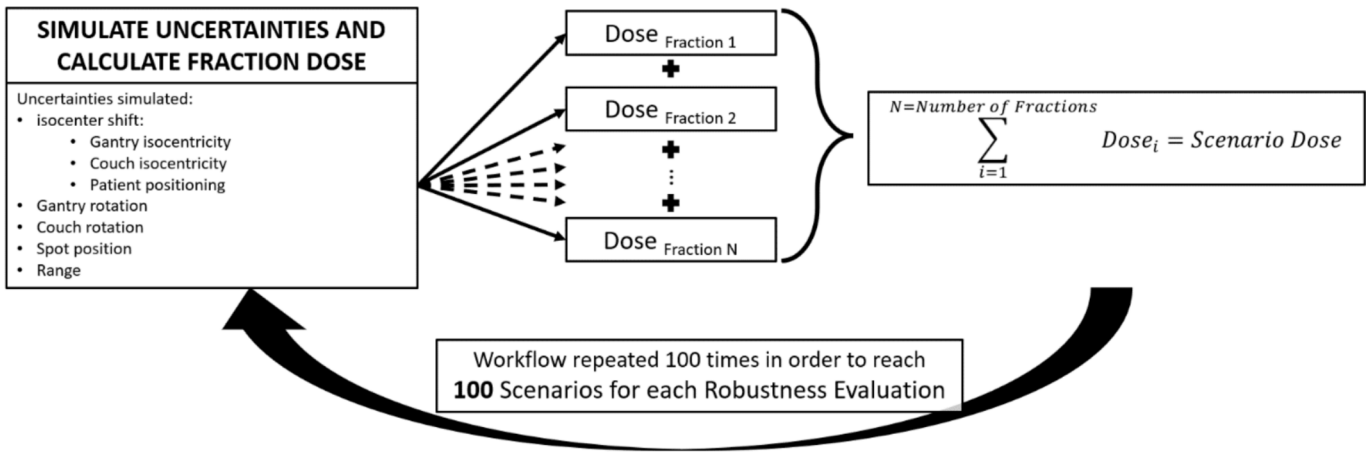


Fig. 1. Flowchart for the probabilistic robustness evaluation. Each fraction dose was simulated by considering all the uncertainties included in this study. The process was repeated N times, where N is the number of fraction of the treatment. The sum of every fraction dose gave the scenario dose. This workflow has been repeated 100 time to decrease statistical uncertainty.

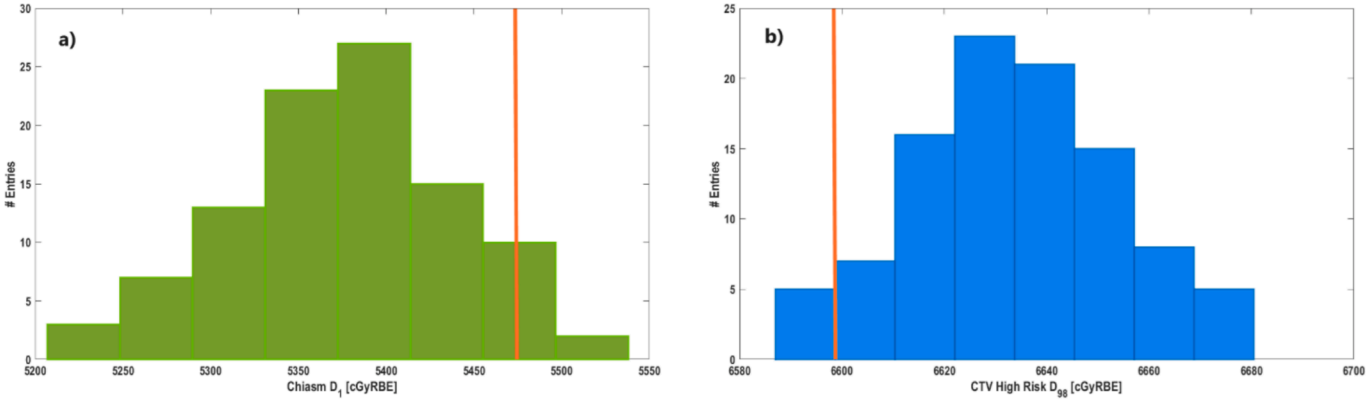


Fig. 2. A) chiasm's d1 distribution from probabilistic robustness evaluation (160 scenarios). the orange line identifies the 95th percentile of all scenarios results. b) ctv d95 distribution from probabilistic robustness evaluation. the orange line identifies the 5th percentile of all scenario results. data from one patient. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

highlighted, respectively. From here on these DVH indexes will be referred to as $D^{th}_{X(95}$ or $D^{th}_{X(5}$ where X will be the percentage of volume of interest, respectively for OARs and for target coverage.

4. Results

4.1. Uncertainty estimation

An example of QA machine data analysis is reported in Fig. 3. Data from spot positioning were summarized as a function of gantry angle and nominal beam energy: along the X direction (in Fig. 3a) and Y direction (in Fig. 3b) in the beam's coordinate system as a function of the gantry angle at 120 MeV are shown. Gantry isocentricity and gantry rotation were analysed as a function of gantry angle (Fig. 3c). The isocentricity uncertainty of the couch is shown in Fig. 3d as a function of couch angle (between −90° and + 90° because of mechanical limits) and of weight load on the couch (in the graph is 60 kg). In all of these graphs the continuous line identifies the mean value averaged through 10 years of data analysis while the coloured area identifies the 1σ region of the data distribution.

All of these data, together with their angular/weight/energy dependences were summarized in lookup-tables that were used by the probabilistic robustness evaluation tool to simulate the correct dependencies. To test the quality of the pseudorandom number generator of our tool, in Fig. 4 we compared the distribution of 10 years of data (histograms in the figure) with the gaussian probability distribution function simulated during robustness evaluations (red curves in the figure). This comparison was performed for gantry rotational uncertainty (shown at gantry 0° in Fig. 4a), for spot positioning (shown at gantry 0° with 100 MeV of nominal beam energy in Fig. 4b), for couch isocentricity and rotational uncertainty (shown at couch 0° with 60 kg of weight on the couch in Fig. 4c and in Fig. 4d), for gantry isocentricity (shown at gantry 0° along the Y axis in Fig. 4e) and for inter-fractional patient positioning uncertainty (shown along Z axis in Fig. 4f).

4.2. Robustness evaluations comparison

Having 100 scenarios simulated with pRE, instead of 16 scenarios (wRE), a new metric needs to be defined in order to interpret and report a large number of DVH indexes and their distribution. The statistical uncertainty analysis of pRE showed that a multiplicity factor of 100 ensured a statistical uncertainty of the percentile value of each dose index studied less than 0.6 % (Supplementary, Table S1).

For each of the OAR selected for this study and for each CTV, $D^{th}_{X(95}$ or $D^{th}_{X(5}$, respectively, were extracted from the DVH bands distribution. They were compared with the worst-case result of the 16 wRE performed for each patient (obtained by combining 4 systematic setup uncertainties – from 0.0 mm to 2.0 mm – with 4 range uncertainties – from 1.0 % to 3.5 %). In Fig. 5 the dose differences in terms of median, 95th and 5th percentile surfaces are shown, both for D_{95} (for target) and D_1 /D_{1cc} (for

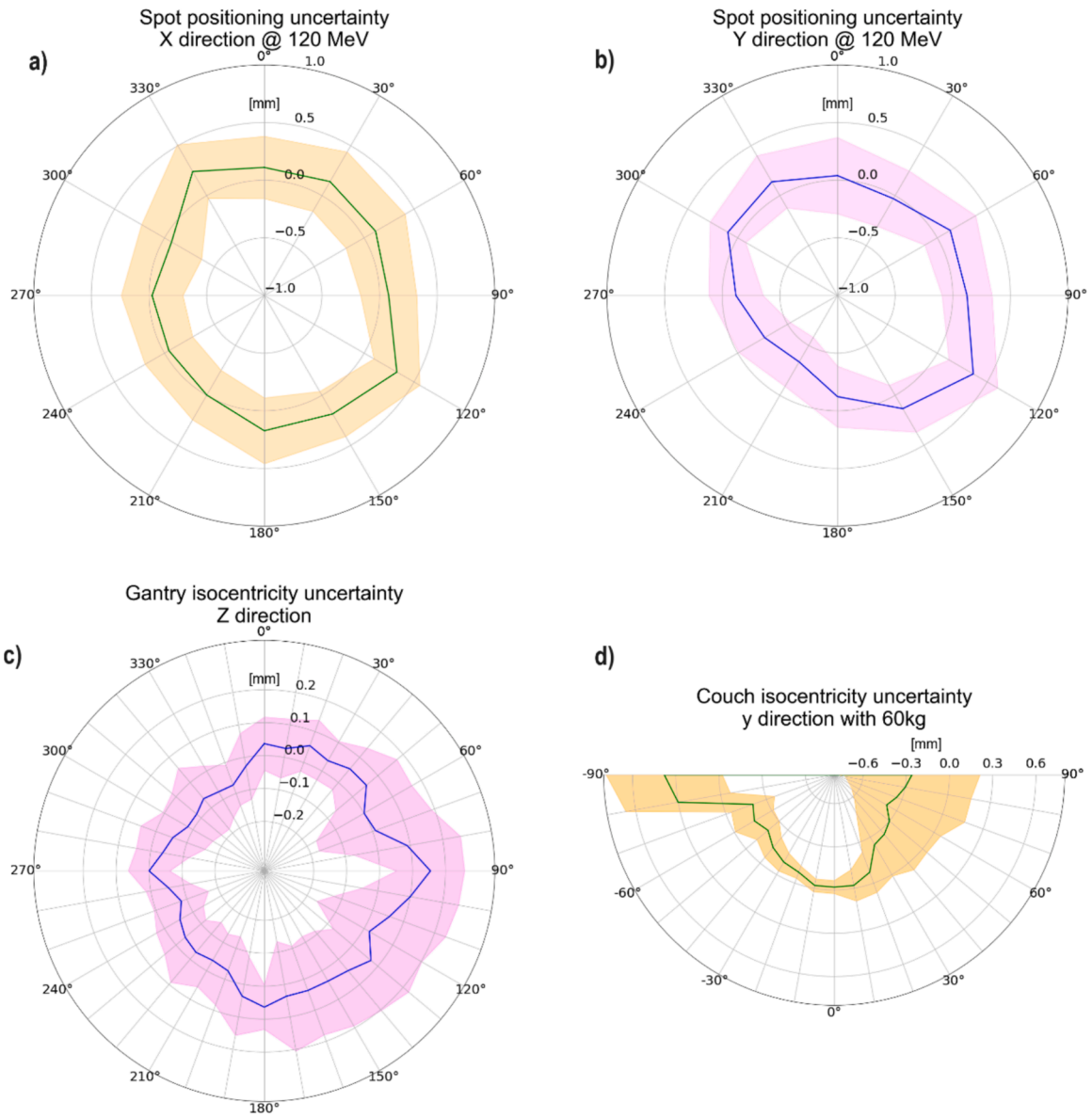


Fig. 3. An example of QA machine data analysis reported here for spot positioning uncertainty (on X and Y direction) at different gantry angle and with a nominal energy of 120 MeV and for gantry c) and couch d) isocentricity uncertainty along z and y axis, respectively. In all of these graphs the continuous line identifies the mean value averaged through 10 years of data analysis while the coloured area identifies the 1σ region of the data distribution.

OARs). The median surface (in Fig. 5 the surface in the middle) represents the median dose differences between pRE and wRE results (the latter performed with the setup and range parameters reported on the x and y-axis). 90 % of the results are within the upper and lower surfaces in the figure. The criteria for choosing the best wRE parameters were defined to have the 95 % of dose differences results *above* 0 GyRBE for OARs and *below* 0 GyRBE for target coverage. With these criteria the risk of having a false negative in wRE analysis is reduced to 5 %. In this case the false negative results can be defined when the D_1 to primary OAR in the worst-case scenario of wRE is lower than the $D^{th}_{1(95}$ of pRE, underestimating the risk of toxicity or, in the same way, when the minimum dose to the target (i.e. D_{95}) in the worst-case scenario of wRE is higher than the $D^{th}_{95(5}$ of pRE, overestimating the tumour dose coverage. A higher risk for OAR (lower for target) would have been deemed clinically unacceptable. The secondary criterion is defined by analysing the median dose difference results aiming at reducing this value as much as possible.

In Fig. 5 the red ovals highlight the best results in terms of both selection criteria. The combination of wRE parameters for setup and range uncertainties that best surrogates the pRE results at the same time for OARs and target coverage is (1.0 mm, 3.5 %): the median DVH indexes differences both for target and OAR reaches very low values (−0.66 and −0.67 GyRBE for D_{98} and D_{95}, respectively, 1.90 GyRBE for OAR's D_1 /D_{1cc}) if compared to wRE parameters with different setup parameter values (2.0, 1.0 mm and 0.0 mm). For OARs, wRE obtained with (1.0 mm,2.0 %) parameters show a lower median value (1.35 GyRBE) than the one obtained with (1.0 mm,3.5 %) and an even lower, but still positive, 5th percentile value (0.13 GyRBE compared to 0.39 GyRBE obtained with 1.0 mm,3.5 % parameters). Despite these results, in order to avoid a differentiation between target and OARs during robustness analysis, (1.0 mm,3.5 %) were selected as representative parameters for wRE that better surrogates pRE results. We reported the detailed analysis for all the comparisons performed between pRE and wRE with 3.5 % of range uncertainties and all the combination of setup uncertainties in Fig. 6. The remaining data not shown here are reported in the supplementary material together with a detailed inter-patient data comparison.

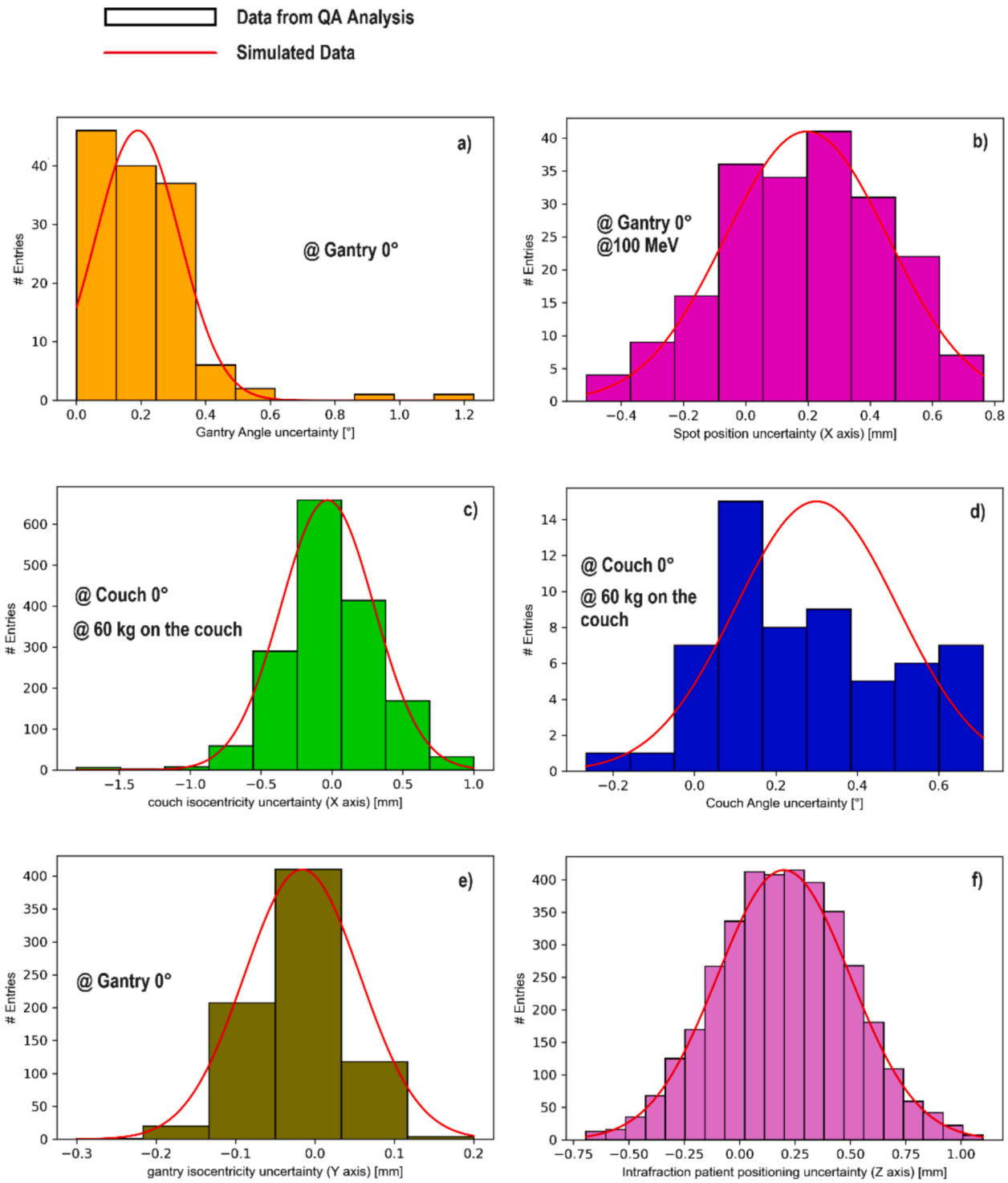


Fig. 4. Comparison of distribution of 10 years of QA data and 157 patient positioning data (histogram) with gaussian probability distribution function simulated during probabilistic robustness evaluations. a) gantry rotational uncertainty here represented for 0° gantry angle; b) spot position uncertainty here represented for data collected at 0° gantry angle and 120 MeV of nominal beam energy; c) couch isocentricity and d) couch rotational uncertainty here represented for couch at 0° with a weight loaded on the couch of 60 kg; e) gantry rotational uncertainty here represented for 0° gantry angle and f) intrafraction patient positioning uncertainty. In each subfigure the axis along which the shift is represented is reported. For the definition of each uncertainty please refer to the body of the manuscript.

4.3. Robustness evaluation time

The study found that pRE took a median time of 18.6 h (SD ± 11.5 h). 11.1 % of the simulation time was needed for dose calculation, 77.3 % for pseudo random number generator sampling from look-up tables (i.e. spot positioning, gantry rotations, setup uncertainties) and 11.6 % for fractionated plan preparation, dose summations and robustness scenarios implementation in the TPS. On the other hand, the wRE simulations took less than 10 min for every patient considered in the study.

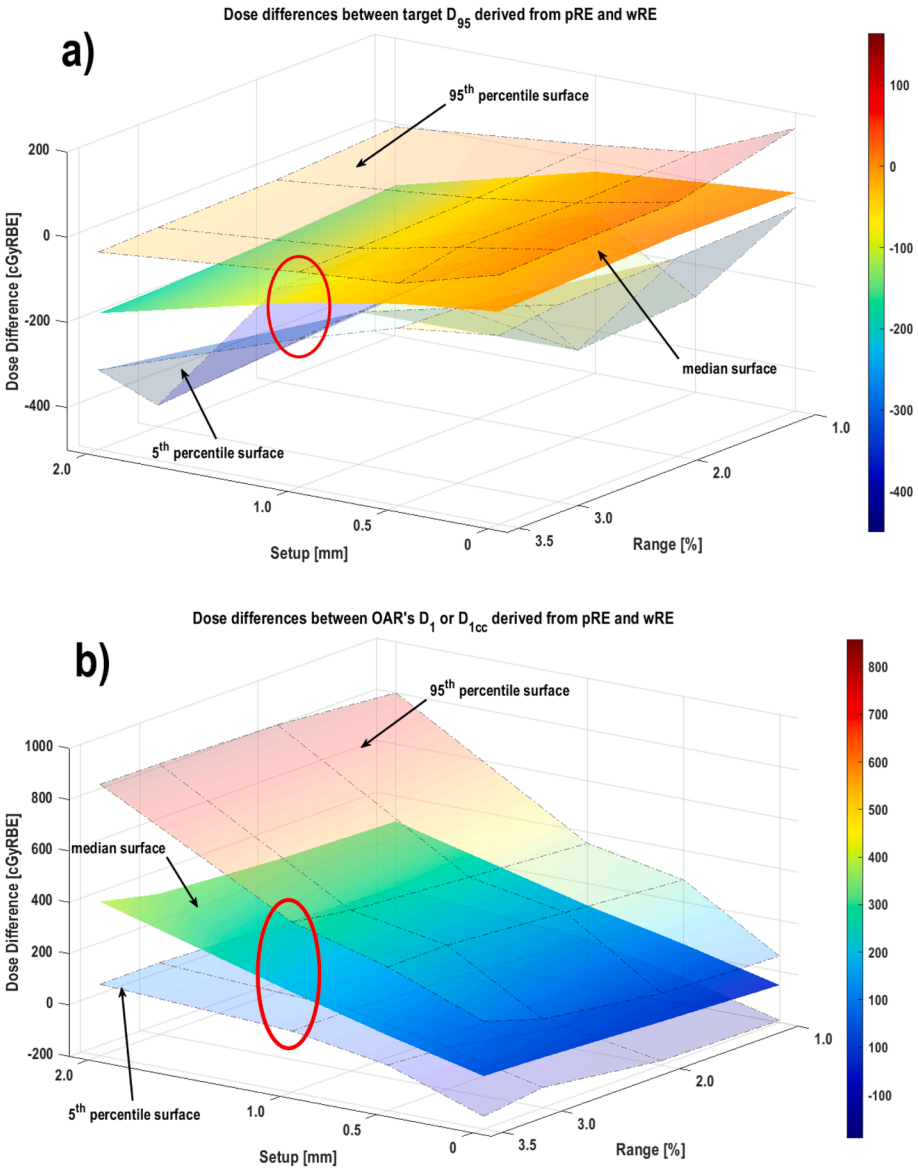


Fig. 5. Dose differences of OARs DVH index of interest (D1 or D1cc) – above – and of target (D95) – below – between pRE and wRE results. On the x and y axis setup and range uncertainties values used in the wRE. The upper and lower surface represent the surface of the 95th and 5th percentile, respectively. The middle surface is the surface of the median dose differences value. The red ovals highlight the combination of parameters that minimize the distances between the three surfaces. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5. Discussion

Extreme caution in choosing the numerical values for the set-up and range errors need to be used in clinical robustness analysis with a worst-case scenario approach. The purpose of this study is to introduce an approach that use data from both machine quality assurance and patient positioning imaging to determine the most effective set of robustness evaluation parameters in commercial treatment planning systems to have a realistic and reliable description of the treatment plan robustness.

To achieve our goal, we needed to establish a standard for evaluating the robustness of our approach. For this purpose, we have developed a new probabilistic method that takes into account all possible uncertainties that may arise during the treatment delivery workflow (anatomy changes, variable RBE dependence are not considered here and will be discussed later in this paragraph). This method is based on a 10-year analysis of machine quality assurance and a large pool of patient positioning data (157). All of these informations have been analysed and used to sample each source of error during the robustness evaluation with a pure Monte Carlo method. The dose calculation was performed for each fraction of the treatment course and each scenario evaluated in the robustness analysis is the result of the sum of each treatment course's fraction dose. The accuracy of dose calculation was granted by the use of a Monte Carlo based dose calculation engine implemented in commercial TPS [24,36]. One of the advantages of this tool is that it is coded in Python: it can be run directly from the TPS user interface and no additional external software/hardware is required. Moreover, it can be shared between centres to test plan robustness considering the real distribution of treatment delivery, geometrical, mechanical and positioning errors.

The combination of a probabilistic approach for robustness assessment and a Monte Carlo dose engine for dose calculation leads, inevitably, to a total average calculation time for a single robustness evaluation that clearly cannot fit into clinical workflows (median calculation time of 18.6 h). For this reason, we have simulated 16 different wRE with different combinations of setup and range values in order to find the combination that best surrogates the pRE results. Each

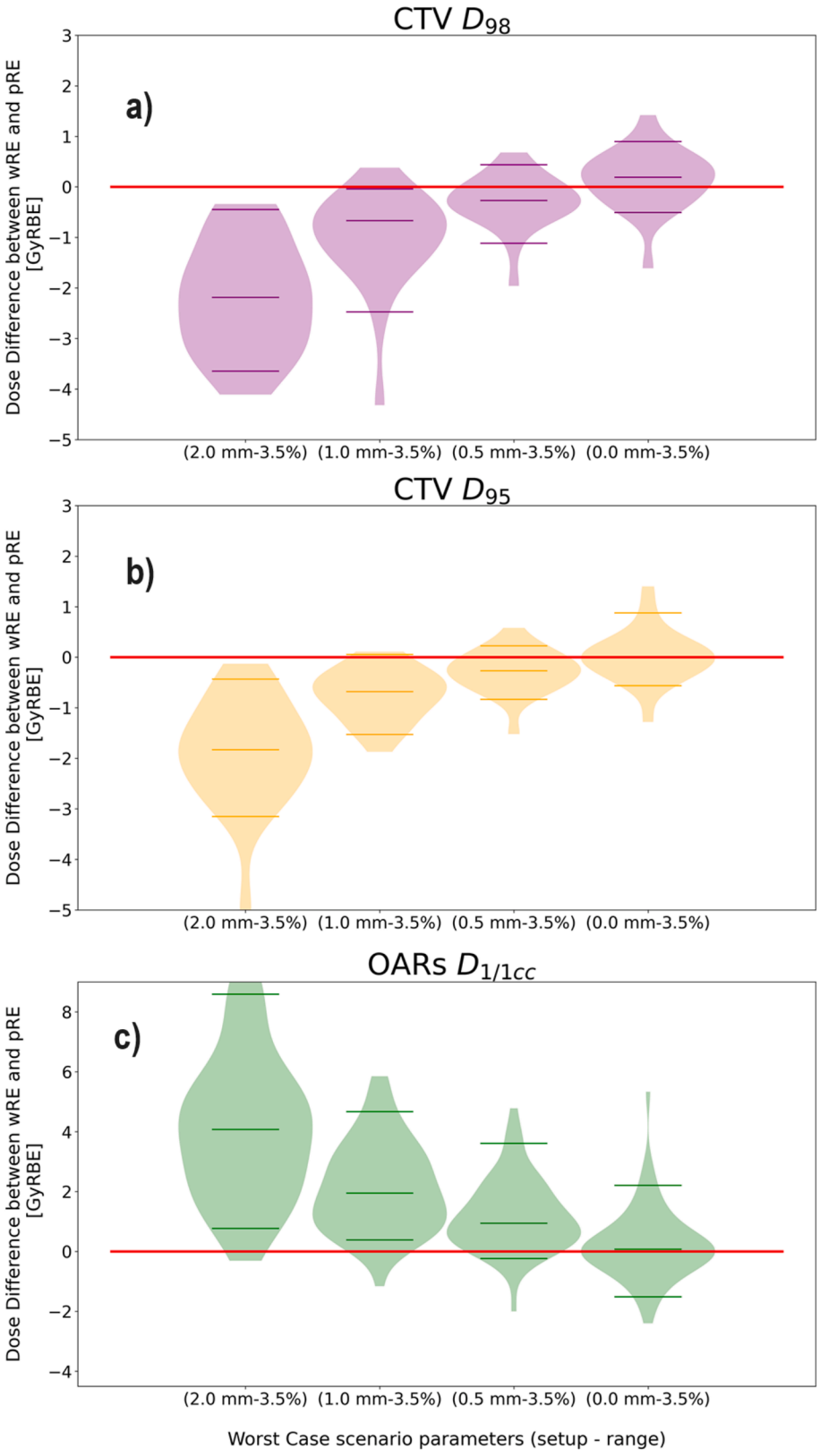


Fig. 6. Dose difference between dose index of interest between pRE and wRE with 3.5% of range uncertainties and all the combination of setup uncertainties. The lines in the violin plot represent the median (in the middle) and the 95th and 5th percentile of each distribution. The red line identifies the zero-dose difference for which data above this number for target and below for OAR can be considered false negatives if wRE is used to surrogate pRE results. The complete analysis is reported in supplementary materials. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

analysis was compared in terms of D_1 or D_{1cc} for the serial organs at risk involved in the analysis and in terms of D_{98} and D_{95} for target coverage. It resulted that the combination of 1.0 mm/3.5 % as systematic setup and range error, respectively, is the one that best surrogates the results of the pRE both in terms of near maximum dose to OARs and in terms of target coverage. The metric to assess the best wRE parameter combination was defined to reduce below 5 % the probability of false negatives and, secondly, to converge as much as possible the median of dose index differences to zero. The choice of 95 % as confidence level in the dosimetric space is restrictive if compared to the commonly used one (90 %) for wRE [5,35]. This was an internal decision made within radiation oncologists and medical physics groups to be as conservative as possible on maximum doses to serial OARs considered in this study. This value can be adapted to clinical decision of each centre. The first aim is guaranteed if, for example, 95th percentile of dose index difference between wRE and pRE for OAR was higher than zero (lower than zero for target dose indexes). Even though wRE obtained with different setup/range parameters have shown a lower median dose difference, the 95th-5th percentile interval was wider and, in part, lower than zero (see Fig. 5). Having such large intervals, the probability of obtaining a D_1/D_{1cc} for OARs from wRE lower than the one obtained with pRE increases until non-clinically acceptable levels (5 %). Moreover, it is worth mentioning that these results highlighted the need to reduce the magnitude of setup errors when the worst-case scenario approach is used in the robustness evaluation of intracranial lesions [5,37,38]. Using parameters such as (2.0 mm, 3.5 %) as setup and range uncertainty in the worst-case scenario approach can lead to an over estimation of the near-max dose to OARs of 4.00 GyRBE (with a [95th;5th] percentile interval of [8.57; 0.78] GyRBE) with respect to a more realistic approach when the real effect of fractionation and of random errors is taken into account.

These results highlighted that probabilistic robustness evaluation results can be predicted with worst-case approaches when the correct set of parameters are used with the great advantage of having a low impact on computation time since each wRE can be run in a few minutes. The small value of systematic setup error (1.0 mm) to be simulated is because the random errors and the fractionation have a strong mitigation effect on the final dose distribution making the range error the main source of nominal plan dose degradation during the treatment.

The strength of the method implemented is that it can be applied to other sites with relatively simple changes (i.e. by studying pre and post treatment images from a pool of patients belonging to that district) with hardware and software used in clinical routine. For example, in our clinical practice, we implemented the same method to find the best wRE parameters to be used for the spine treatments. The other advantage is that this method can be considered a reference in the future to test other robustness evaluation methods (i.e. probabilistic approaches in commercial TPS).

The study also has some weaknesses. First, not all the data from the QA analysis have the same numerosity: some QA were performed during the commissioning phase and they were not repeated during the clinical activity (i.e. the test for gantry and couch rotation uncertainty). Nonetheless, the vast majority of the tests had a very large amount of data as shown in Fig. 3 and Fig. 4. The hypothesis that every error was distributed with a Gaussian shaped probability distribution function is an a priori assumption that can be considered valid because of the central limit theorem: the combination of different and independent variables will always be normally distributed regardless of whether their original probability distribution is. Another weakness is related to the number of patients included in the study (12). This number can be increased to obtain more statistically significant results. Nonetheless, even if the number of patients is small we have analyzed 60 OAR's dose indexes and 24 target's dose indexes increasing the statistical significance of the population analyzed.

Some uncertainties were not considered in this analysis like anatomy changes, contouring and radiobiology uncertainty. One of the reasons why these uncertainties were not included was because the plans were generated by using robust optimization approach in which only setup and range uncertainty were considered and none of the above-mentioned uncertainties were included in the optimization process. For consistency with the planning approach, they were not evaluated in the method proposed here. At the time being, this method is able to test the plan robustness only for the above-mentioned machine related uncertainties. All the other uncertainties must be evaluated and, in some form, limited with a proper definition of clinical workflows. In our centre, for example, we have defined a workflow for a) the choice of beams ballistics that implicitly takes into account the possible variation of RBE over the beam's path, b) the cross check of patient's contouring by a second clinician and, c) for patients that have a high probability of anatomy change, the acquisition of repeated CT in treatment position over the treatment course (by using a CT on Rail located in the treatment room) to evaluate if some anatomy changes occurred and their impact on the dose distribution. Moreover, contouring uncertainty is further reduced by using multimodality imaging (MR and PET) for target delineation and automated delineation tools such as deep learning segmentation for organs at risk [39]. Given all of these considerations and the results of this analysis, in our clinical workflow we use 1.0 mm and 3.5 % as parameters for setup and range uncertainties in wRE of skull base chordoma cases.

6. Conclusion

Our results showed that there must be extreme caution in choosing the values for the set-up errors to be used in the robustness analysis with a worst-case approach. Numbers that may even seem reasonable, such as 2–3 mm for brain patients, can lead to unrealistic OAR and target dosimetric indexes. Since these robustness analysis tools have been part of the clinical workflow for some years now, it is crucial that they should be carefully validated given their potential impact on clinical choices.

We developed a tool able to simulate, with a probabilistic approach, the effect of the uncertainties that can occur during the proton therapy treatment and the mitigation effect of the fractionation on nominal dose in order to have a realistic evaluation of plan robustness. Every proton centre in line of principle should perform such kind of analysis based on the value of their uncertainties derived from patient positioning workflow and QA data review.

Since the simulation time of such evaluations does not fit within clinical time workflows, results were compared with a classic worst-case approach implemented in commercial TPS to find the best uncertainty parameters that were able to surrogate the probabilistic approach results. The parameters determined with this study are now used in clinical practice for treatment plan approval.

7. Ethical procedure

The research meets all applicable standards with regard to the ethics of experimentation and research integrity, and the following is being certified/declared true.

As an expert scientist and along with co-authors of concerned field, the paper has been submitted with full responsibility, following due ethical procedure, and there is no duplicate publication, fraud, plagiarism.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

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References

[1] Paganetti H. Range uncertainties in proton therapy and the role of Monte Carlo simulations. Phys Med Biol 2012;57:R99–. https://doi.org/10.1088/0031-9155/57/11/R99.

[2] McNamara A, Willers H, Paganetti H. Modelling variable proton relative biological effectiveness for treatment planning. Br J Radiol 2020;93:20190334. https://doi.org/10.1259/bjr.20190334.

[3] Mohan R, Bortfeld T. Proton therapy: clinical gains through current and future treatment programs. Front Radiat Ther Oncol 2011;43:440–64. https://doi.org/10.1159/000322509.

[4] Kaplan L, Placidi L, Bäck A, Canters R, Hussein M, Vaniqui A, et al. Plan quality assessment in clinical practice: Results of the 2020 ESTRO survey on plan complexity and robustness. Radiother Oncol J Eur Soc Ther Radiol Oncol 2022;173. 10.1016/j.radonc.2022.06.005.

[5] Sterpin E, Widesott L, Poels K, Hoogeman M, Korevaar E, Lowe M, et al. Robustness evaluation of pencil beam scanning proton therapy treatment planning: A systematic review. Radiother Oncol J Eur Soc Ther Radiol Oncol 2024;197. 10.1016/j.radonc.2024.110365.

[6] Goddard LC, Brodin NP, Bodner WR, Garg MK, Tomé WA. Comparing photon and proton-based hypofractioned SBRT for prostate cancer accounting for robustness and realistic treatment deliverability. Br J Radiol 2018;91:20180010. https://doi.org/10.1259/bjr.20180010.

[7] Liu C, Yu NY, Shan J, Bhangoo RS, Daniels TB, Chiang JS, et al. Technical Note: Treatment planning system (TPS) approximations matter - comparing intensity-modulated proton therapy (IMPT) plan quality and robustness between a commercial and an in-house developed TPS for nonsmall cell lung cancer (NSCLC). Med Phys 2019;46:4755–62. https://doi.org/10.1002/mp.13809.

[8] Butkus MP, Brovold N, Diwanji T, Xu Y, De Ornelas M, Dal Pra A, et al. Assessment of IMPT versus VMAT plans using different uncertainty scenarios for prostate cancer. Radiat Oncol Lond Engl 2022;17:162. https://doi.org/10.1186/s13014-022-02126-y.

[9] Guerreiro F, Zachiu C, Seravalli E, Ribeiro CO, Janssens GO, Ries M, et al. Evaluating the benefit of PBS vs. VMAT dose distributions in terms of dosimetric sparing and robustness against inter-fraction anatomical changes for pediatric abdominal tumors. Radiother Oncol J Eur Soc Ther Radiol. Oncol 2019;138:158–65. https://doi.org/10.1016/j.radonc.2019.06.025.

[10] Gutierrez A, Rompokos V, Li K, Gillies C, D'Souza D, Solda F, et al. The impact of proton LET/RBE modeling and robustness analysis on base-of-skull and pediatric craniopharyngioma proton plans relative to VMAT. Acta Oncol Stockh Swed 2019;58:1765–74. https://doi.org/10.1080/0284186X.2019.1653496.

[11] Hoffmann L, Mortensen H, Shamshad M, Berbee M, Bizzocchi N, Bütof R, et al. Treatment planning comparison in the PROTECT-trial randomising proton versus photon beam therapy in oesophageal cancer: Results from eight European centres. Radiother Oncol J Eur Soc Ther Radiol Oncol 2022;172:32–41. https://doi.org/10.1016/j.radonc.2022.04.029.

[12] Tommasino F, Widesott L, Fracchiolla F, Lorentini S, Righetto R, Algranati C, et al. Clinical implementation in proton therapy of multi-field optimization by a hybrid method combining conventional PTV with robust optimization. Phys Med Biol 2020;65:045002. https://doi.org/10.1088/1361-6560/ab63b9.

[13] Bodensteiner D. RayStation: External beam treatment planning system. Med Dosim Off J Am Assoc Med Dosim 2018;43:168–76. https://doi.org/10.1016/j.meddos.2018.02.013.

[14] Wahl N, Hennig P, Wieser H, Bangert M. Analytical probabilistic modeling of dose-volume histograms. Med Phys 2020;47. https://doi.org/10.1002/mp.14414.

[15] Wahl N, Hennig P, Wieser H-P, Bangert M. Analytical incorporation of fractionation effects in probabilistic treatment planning for intensity-modulated proton therapy. Med Phys 2018;45:1317–28. https://doi.org/10.1002/mp.12775.

[16] Wahl N, Hennig P, Wieser HP, Bangert M. Efficiency of analytical and sampling-based uncertainty propagation in intensity-modulated proton therapy. Phys Med Biol 2017;62:5790–807. https://doi.org/10.1088/1361-6560/aa6ec5.

[17] Souris K, Barragan Montero A, Janssens G, Di Perri D, Sterpin E, Lee JA. Technical Note: Monte Carlo methods to comprehensively evaluate the robustness of 4D treatments in proton therapy. Med Phys 2019;46:4676–84. https://doi.org/10.1002/mp.13749.

[18] Perkó Z, van der Voort SR, van de Water S, Hartman CMH, Hoogeman M, Lathouwers D. Fast and accurate sensitivity analysis of IMPT treatment plans using Polynomial Chaos Expansion. Phys Med Biol 2016;61:4646–64. https://doi.org/10.1088/0031-9155/61/12/4646.

[19] Rojo-Santiago J, Korevaar E, Perkó Z, Both S, Habraken SJM, Hoogeman MS. PTV-based VMAT vs. robust IMPT for head-and-neck cancer: A probabilistic uncertainty analysis of clinical plan evaluation with the Dutch model-based selection. Radiother Oncol J Eur Soc Ther Radiol Oncol 2023;186:109729. 10.1016/j.radonc.2023.109729.

[20] Sterpin E, Rivas ST, Van den Heuvel F, George B, Lee JA, Souris K. Development of robustness evaluation strategies for enabling statistically consistent reporting. Phys Med Biol 2021;66:045002. https://doi.org/10.1088/1361-6560/abd22f.

[21] Stammer P, Burigo L, Jäkel O, Frank M, Wahl N. Efficient uncertainty quantification for Monte Carlo dose calculations using importance (re-)weighting. Phys Med Biol 2021;66. https://doi.org/10.1088/1361-6560/ac287f.

[22] Vazquez I, Gronberg MP, Zhang X, Court LE, Zhu XR, Frank SJ, et al. A deep learning-based approach for statistical robustness evaluation in proton therapy treatment planning: a feasibility study. Phys Med Biol 2023;68. https://doi.org/10.1088/1361-6560/accc08.

[23] Fracchiolla F, Fellin F, Innocenzi M, Lipparini M, Lorentini S, Widesott L, et al. A pre-absorber optimization technique for pencil beam scanning proton therapy treatments. Phys Med 2019;57. https://doi.org/10.1016/j.ejmp.2018.12.014.

[24] Fracchiolla F, Engwall E, Janson M, Tamm F, Lorentini S, Fellin F, et al. Clinical validation of a GPU-based Monte Carlo dose engine of a commercial treatment planning system for pencil beam scanning proton therapy. Phys Medica PM Int J Devoted Appl Phys Med Biol Off J Ital Assoc Biomed Phys AIFB 2021;88. https://doi.org/10.1016/j.ejmp.2021.07.012.

[25] Depuydt T, Penne R, Verellen D, Hrbacek J, Lang S, Leysen K, et al. Computer-aided analysis of star shot films for high-accuracy radiation therapy treatment units. Phys Med Biol 2012;57. https://doi.org/10.1088/0031-9155/57/10/2997.

[26] Bizzocchi N, Fracchiolla F, Schwarz M, Algranati C. A fast and reliable method for daily quality assurance in spot scanning proton therapy with a compact and inexpensive phantom. Med Dosim Off J Am Assoc. Med Dosim 2017;42. https://doi.org/10.1016/j.meddos.2017.05.001.

[27] Arjomandy B, Taylor P, Ainsley C, Safai S, Sahoo N, Pankuch M, et al. AAPM task group 224: Comprehensive proton therapy machine quality assurance. Med Phys 2019;46:e678–705. https://doi.org/10.1002/mp.13622.

[28] Grevillot L, Moreno JO, Fuchs H, Dreindl R, Elia A, Bolsa-Ferruz M, et al. Implementation of Sphinx/Lynx as daily QA equipment for scanned proton and carbon ion beams. J Appl Clin Med Phys 2023;24:e13896. https://doi.org/10.1002/acm2.13896.

[29] Jiang H, Seco J, Paganetti H. Effects of Hounsfield number conversion on CT based proton Monte Carlo dose calculations - PubMed. Med Phys n.d.

[30] Peters N, Wohlfahrt P, Dahlgren C, de Marzi L, Ellerbrock M, Fracchiolla F, et al. Experimental assessment of inter-centre variation in stopping-power and range prediction in particle therapy. Radiother Oncol J Eur Soc Ther Radiol Oncol 2021;163. 10.1016/j.radonc.2021.07.019.

[31] Peters N, Wohlfahrt P, Hofmann C, Möhler C, Menkel S, Tschiche M, et al. Reduction of clinical safety margins in proton therapy enabled by the clinical implementation of dual-energy CT for direct stopping-power prediction. Radiother Oncol J Eur Soc Ther Radiol Oncol 2022;166:71–8. https://doi.org/10.1016/j.radonc.2021.11.002.

[32] Cohilis M, Souris K, Buti G, Chang C, Lin L, Lee J, et al. A spot-specific range uncertainty framework for robust optimization of proton therapy treatments. Med Phys 2023;50. https://doi.org/10.1002/mp.16706.

[33] Farace P, Tommasino F, Righetto R, Fracchiolla F, Scaringella M, Bruzzi M, et al. Technical Note: CT calibration for proton treatment planning by cross-calibration with proton CT data. Med Phys 2021;48:1349–55. https://doi.org/10.1002/mp.14698.

[34] Lowe M, Gosling A, Nicholas O, Underwood T, Miles E, Chang Y, et al. Comparing Proton to Photon Radiotherapy Plans: UK Consensus Guidance for Reporting Under Uncertainty for Clinical Trials. Clin Oncol R Coll Radiol G B 2020;32. https://doi.org/10.1016/j.clon.2020.03.014.

[35] Korevaar EW, Habraken SJM, Scandurra D, Kierkels RGJ, Unipan M, Eenink MGC, et al. Practical robustness evaluation in radiotherapy - A photon and proton-proof alternative to PTV-based plan evaluation. Radiother Oncol J Eur Soc Ther Radiol Oncol 2019;141:267–74. https://doi.org/10.1016/j.radonc.2019.08.005.

[36] Widesott L, Lorentini S, Fracchiolla F, Farace P, Schwarz M. Improvements in pencil beam scanning proton therapy dose calculation accuracy in brain tumor cases with a commercial Monte Carlo algorithm. Phys Med Biol 2018;63. https://doi.org/10.1088/1361-6560/aac279.

[37] Noufal MP, Widesott L, Sharma SD, Righetto R, Cianchetti M, Schwarz M. The Role of Plan Robustness Evaluation in Comparing Protons and Photons Plans - An Application on IMPT and IMRT Plans in Skull Base Chordomas. J Med Phys 2020;45:206–14. https://doi.org/10.4103/jmp.JMP_45_20.

[38] Argota-Perez R, Sharma MB, Elstrøm UV, Møller DS, Grau C, Jensen K, et al. Dose and robustness comparison of nominal, daily and accumulated doses for photon and proton treatment of sinonasal cancer. Radiother Oncol J Eur Soc Ther Radiol Oncol 2022;173:102–8. https://doi.org/10.1016/j.radonc.2022.05.038.

[39] Alzahrani N, Henry A, Clark A, Murray L, Nix M, Al-Qaisieh B. Geometric evaluations of CT and MRI based deep learning segmentation for brain OARs in radiotherapy. Phys Med Biol 2023;68. https://doi.org/10.1088/1361-6560/acf023.